ADVANCED MACHINE LEARNING · WORKSHEET 16

Decision Trees Part I

Learned rules and CART classification and regression

STUDENT COPY

Name	Roll No.	Date

1 Opening scenario predicting student placement

Continue the NST student-placement story from Logistic Regression, but use the same two input features that students already know from the earlier regression examples: **Hours studied** and **Attendance percentage**. The earlier continuous **Score** column is deliberately removed. Today the target is binary: **Placement** y, where 1 means placed and 0 means not placed.

Let's try to do the classification using normal if–else statements.

Use this **six-row NST board dataset** throughout Lecture 16, the paper worksheet and the manual-split warm-up in Lab 16. The small size is intentional: every candidate split can be checked by hand.

Student	Hours studied	Attendance (%)	Placement y
1	2	60	0
2	4	80	1
3	6	90	1
4	5	50	0
5	7	68	0
6	3	85	0
New student	7	82	?

Classroom prompt: Instead of one straight boundary, what if the rule were "IF Attendance $> 75\%$, then ask whether Hours studied > 3.5"? Is that still a valid model?

Expected response: Yes. It is a nested if–else program. A decision tree learns such questions from data rather than using a weighted sum of the features. We can visualize the questions as a flowchart ending in a prediction.

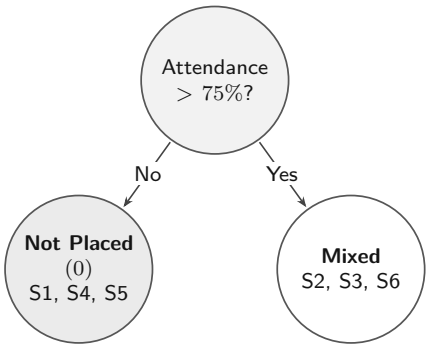
KEY INSIGHT

The opening uses only if–else routing. The two thresholds illustrate the rule; later CART will compare candidates and learn split choices from these observations.

1A First condition IF Attendance exceeds 75 percent

Apply only **Attendance** $> 75\%$**?** Light gray highlights the training rows taking **Yes**: S2, S3 and S6. The new student has no training label and stays separate.

Student	Hours studied	Attendance (%)	Placement y
S1	2	60	0
S2	4	80	1
S3	6	90	1
S4	5	50	0
S5	7	68	0
S6	3	85	0
New student	7	82	?



The **No** branch contains S1, S4 and S5, all with target 0, so it becomes a Not Placed leaf. The **Yes** branch contains S2 and S3 with target 1 and S6 with target 0. It remains mixed and needs another question.

TAKEAWAY

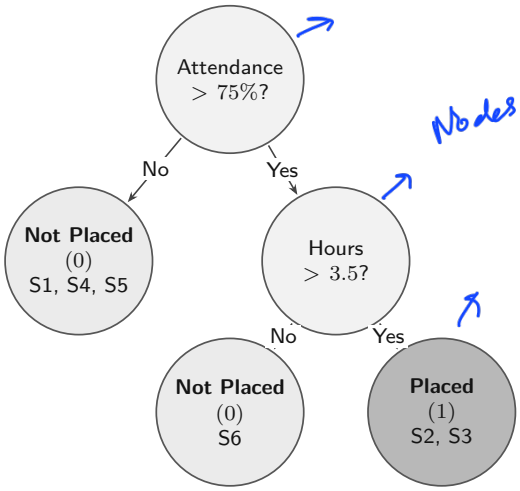
Answering Yes to an input question does not automatically mean that the target is 1. S6 has Attendance above 75% and is still Not Placed.

1B Complete the tree then read its prediction regions

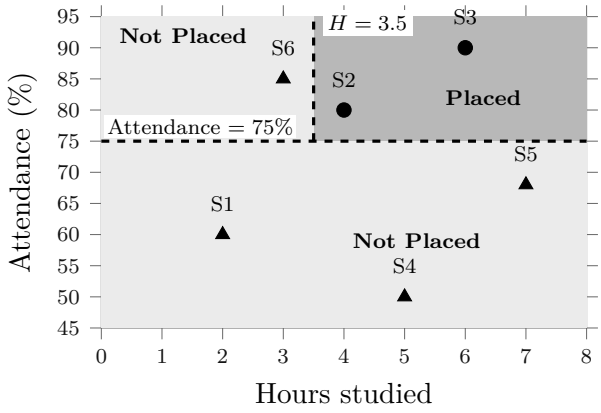
Now ask **whether Hours studied** > 3.5, only within the first Yes branch. S2 and S3 satisfy both conditions and become **dark gray**. S6 remains light gray: it passed Attendance, then answered No to Hours. S1/S4/S5 never receive the second question.

Student	Hours studied	Attendance (%)	Placement y
S1	2	60	0
S2	4	80	1
S3	6	90	1
S4	5	50	0
S5	7	68	0
S6	3	85	0
New student	7	82	?

Both splits together



The same rows in feature space



The root is at depth 0; the Hours question is at depth 1, and its leaves are at depth 2.

The original No branch is already a leaf at depth 1. The new Hours node is at depth 1; its two leaves are at depth 2. All six training students now reach a pure leaf.

TAKEAWAY

Dark rows S2/S3 reach Placed (1). The light row S6 reaches Not Placed (0). The first No rows S1/S4/S5 also reach Not Placed (0).

Keep the horizontal boundary at Attendance = 75%. Draw the vertical boundary at Hours = 3.5 **only above Attendance 75**: the second question belongs to that branch, not to the whole dataset.

● Placed ($y = 1$) ▲ Not Placed ($y = 0$)

Two sequential axis-parallel splits form three rectangular prediction regions.

Region 1: Attendance $\leq 75\%$ $\rightarrow$ Not Placed; this contains S1, S4 and S5.

Region 2: Attendance $> 75\%$ and Hours ≤ 3.5 $\rightarrow$ Not Placed; this contains S6.

Region 3: Attendance $> 75\%$ and Hours > 3.5 $\rightarrow$ Placed; this contains S2 and S3.

New student $(7, 82)$**:** $82 > 75$, then $7 > 3.5$. The path is Yes $\rightarrow$ Yes $\rightarrow$ Placed (1), and the point lies in the darker grey Placed region.

KEY INSIGHT

A later split changes only its own branch. The table shading, the leaf memberships and the three scatter regions all express the same nested rule.

2 What is a Decision Tree and why do we need it

HOOK

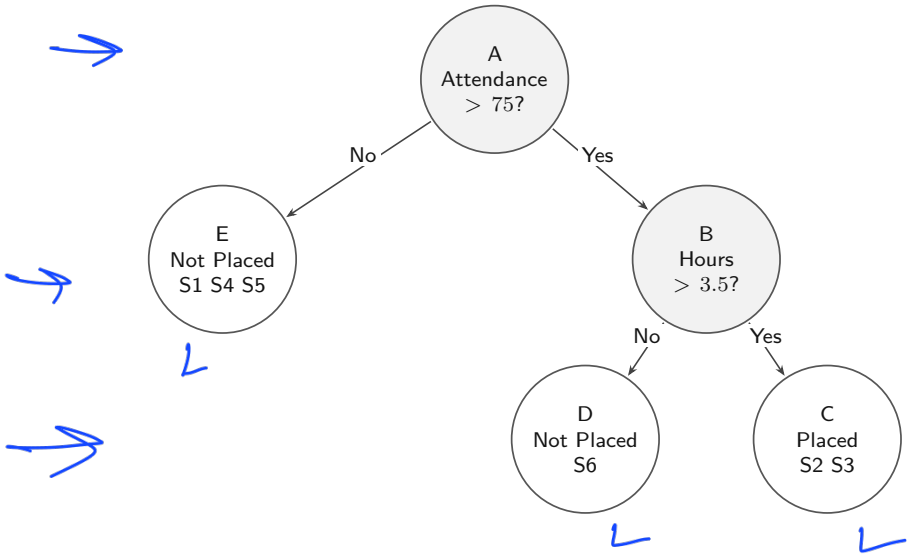
The NST rule first checks Attendance and asks about Hours only in its Yes branch. A tree lets one question depend on the answers to earlier questions; we can now name this structure.

A **decision tree** is a supervised model that repeatedly partitions the input data. Each internal node asks a feature question, each branch represents an answer, and each leaf supplies a prediction.

For classification, a leaf predicts a class, commonly the training majority in that leaf. For regression, it predicts a numerical value. We will use CART to develop both tasks.

Non-parametric means the tree is not restricted to a fixed-size coefficient vector or one prescribed functional shape. A fitted tree still has learned thresholds, leaf values and training controls. Small trees give readable rules; very deep trees are harder to interpret.

2A Reading the tree structure terminology



Read the same complete opening rule with node labels A–E. The candidate thresholds have not yet been selected by comparing split scores.

Term	Meaning in the NST tree
Root	A: the first Attendance question
Decision node	B: the Hours question
Branch	A Yes or No answer that routes the row
Leaf	C,D,E: terminal predictions
Parent and child	A is a parent of B; B is a child of A
Subtree	B together with C,D and the connecting edges
Depth	Edges from the root: A=0, B/E=1, C/D=2

3 CART classification start with rows and features

CART = Classification and Regression Trees. It is an algorithm for growing decision trees.

HOOK

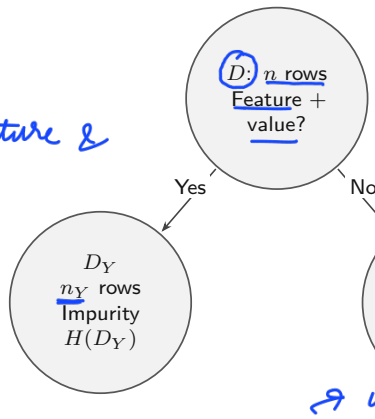
Let's take an example of a dataset in which you have n rows and m features/columns. You have the last column as the categorical target, having the values Yes (Y) or No (N).

The feature columns contain the information used to ask questions. The last column tells us the observed class; it supplies the labels used to assess the split.

f_1	f_2	$\cdots$	f_m	**Category**
x_{11}	x_{12}	$\cdots$	x_{1m}	Y
x_{21}	x_{22}	$\cdots$	x_{2m}	N
x_{31}	x_{32}	$\cdots$	x_{3m}	Y
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
x_{n1}	x_{n2}	$\cdots$	x_{nm}	N

To make a decision node, we need two things: **a feature and a value**. The feature tells us which column to use. The value completes the question: for example, Field = Science or Grade > 85.5.

How do we choose these two things? Try the valid feature–value questions. For each trial, divide the current data into two groups, measure their impurity, and compare their weighted score.



$$J = \frac{n_Y}{n} H(D_Y) + \frac{n_N}{n} H(D_N), \qquad n_Y + n_N = n.$$

The weights count how much of the current dataset reaches each child. Choose the question with the lowest weighted impurity, then repeat the same process separately inside each child.

TAKEAWAY

A Yes branch is an answer to an input question. It is not automatically the target class Y. Each child can still contain both target classes.

3A Employment data and the classification criterion

Consider this ten-row employment dataset. Degree and Field are categorical inputs; Average Grade is numerical. Job Outcome is the target: E = Employed and U = Unemployed. In the generic target description, E corresponds to Yes (Y) and U to No (N). Answers to an input question are separate from these target labels.

Row	Degree	Field	Average Grade	Job Outcome
0	UG	Science	89	E
1	UG	Arts	92	U
2	PG	Science	95	E
3	PhD	Science	85	E
4	PG	Arts	98	U
5	PhD	Arts	90	E
6	UG	Science	88	U
7	PG	Arts	93	E
8	UG	Arts	94	U
9	PhD	Science	86	E

We will use Gini as the classification impurity function:

$$G(D) = 1 - \sum_{k=1}^{K} p_k^2.$$

Here p_k is the proportion of target class k among the rows reaching this node. There are two classes, so $G = 1 - p_E^2 - p_U^2$. In the complete dataset, six rows are E and four are U:

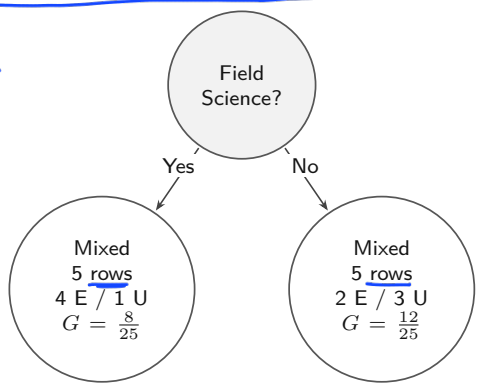
$$p_E = 6/10, \quad p_U = 4/10, \quad G(D) = 1 - (6/10)^2 - (4/10)^2 = 12/25 = 0.48.$$

3B Categorical example Field equals Science

Test this question on the original dataset D: rows 0–9, with 6 E and 4 U. The parent Gini is $12/25 = 0.48$. This is an alternative root trial, not an added level of the tree.

Branch	Rows and observed labels	Count
Yes	0 E, 2 E, 3 E, 6 U, 9 E	5: 4 E / 1 U
No	1 U, 4 U, 5 E, 7 E, 8 U	5: 2 E / 3 U

$$G_Y = 1 - (\tfrac{4}{5})^2 - (\tfrac{1}{5})^2 = \frac{8}{25} = 0.3200,$$
$$G_N = 1 - (\tfrac{2}{5})^2 - (\tfrac{3}{5})^2 = \frac{12}{25} = 0.4800,$$
$$J_G = \frac{5}{10}(\frac{8}{25}) + \frac{5}{10}(\frac{12}{25}) = \frac{2}{5} = 0.4000,$$
$$\Delta G = \frac{12}{25} - \frac{2}{5} = \frac{2}{25} = 0.0800.$$



This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

TAKEAWAY

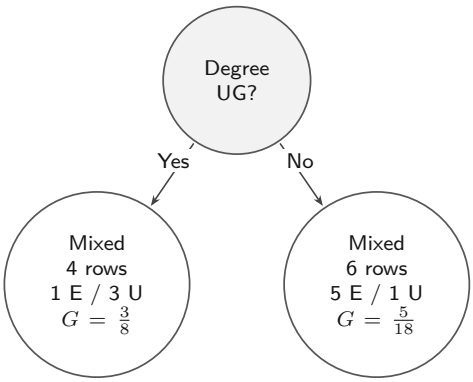
Field has only Science and Arts. Asking Field = Arts creates the same two children in reversed order, so its weighted Gini is also 0.4. A second full calculation is unnecessary.

3B Categorical example Degree equals UG

Test this question on the original dataset D: rows 0–9, with 6 E and 4 U. The parent Gini is $12/25 = 0.48$. This is an alternative root trial, not an added level of the tree.

Branch	Rows and observed labels	Count
Yes	0 E, 1 U, 6 U, 8 U	4: 1 E / 3 U
No	2 E, 3 E, 4 U, 5 E, 7 E, 9 E	6: 5 E / 1 U

$$G_Y = 1 - (\tfrac{1}{4})^2 - (\tfrac{3}{4})^2 = \frac{3}{8} = 0.3750,$$
$$G_N = 1 - (\tfrac{5}{6})^2 - (\tfrac{1}{6})^2 = \frac{5}{18} = 0.2778,$$
$$J_G = \frac{4}{10}(\frac{3}{8}) + \frac{6}{10}(\frac{5}{18}) = \frac{19}{60} = 0.3167,$$
$$\Delta G = \frac{12}{25} - \frac{19}{60} = \frac{49}{300} = 0.1633.$$



This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

The other questions under Degree are scored in the same way. Their final weighted child Gini values are:

Degree question	Weighted child Gini
Degree = PG	$10/21 = 0.4762$
Degree = PhD	$12/35 = 0.3429$

3C Numerical inputs sort Grade and generate thresholds

For Average Grade, arrange the rows in ascending feature order and retain each row's target label. Each adjacent pair of distinct grades gives one midpoint candidate. Equal numerical values would stay together.

Row	Sorted Grade	Outcome	Midpoint to the next value
3	85	E	$(85 + 86)/2 = 85.5$
9	86	E	$(86 + 88)/2 = 87$
6	88	U	$(88 + 89)/2 = 88.5$
0	89	E	$(89 + 90)/2 = 89.5$
5	90	E	$(90 + 92)/2 = 91$
1	92	U	$(92 + 93)/2 = 92.5$
7	93	E	$(93 + 94)/2 = 93.5$
8	94	U	$(94 + 95)/2 = 94.5$
2	95	E	$(95 + 98)/2 = 96.5$
4	98	U	No next value

The nine trial thresholds are 85.5, 87, 88.5, 89.5, 91, 92.5, 93.5, 94.5 and 96.5. Test each midpoint with the question:

$$\text{Grade} > t? \qquad \text{Yes: Grade} > t, \quad \text{No: Grade} \le t.$$

The first worked threshold is $t = 85.5$; the next example tests $t = 87$, and so on. Each trial starts again with all ten rows. Sorting proposes questions; the target counts determine how good each question is. Keep all ten observations in the sorted table. The largest value, 98, gives the final midpoint $(95 + 98)/2 = 96.5$.

TAKEAWAY

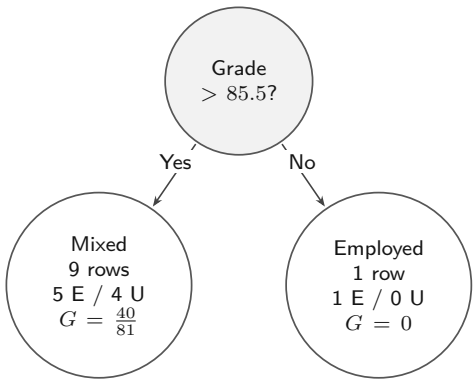
A midpoint lies between observed values. Its comparison routes whole rows into children; it does not average the target labels.

3C Numerical example Grade greater than 85.5

Test this question on the original dataset D: rows 0–9, with 6 E and 4 U. The parent Gini is $12/25 = 0.48$. This is an alternative root trial, not an added level of the tree.

Branch	Rows and observed labels	Count
Yes	0 E, 1 U, 2 E, 4 U, 5 E, 6 U, 7 E, 8 U, 9 E	9: 5 E / 4 U
No	3 E	1: 1 E / 0 U

$$G_Y = 1 - (\frac{5}{9})^2 - (\frac{4}{9})^2 = \frac{40}{81} = 0.4938,$$
$$G_N = 1 - (\frac{1}{1})^2 - (\frac{0}{1})^2 = 0 = 0.0000,$$
$$J_G = \frac{9}{10}(\frac{40}{81}) + \frac{1}{10}(0) = \frac{4}{9} = 0.4444$$
$$\Delta G = \frac{12}{25} - \frac{4}{9} = \frac{8}{225} = 0.0356.$$



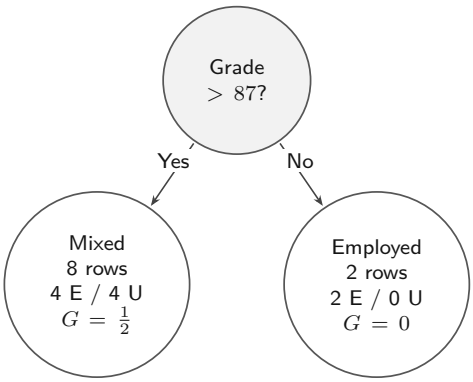
This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

3C Numerical example Grade greater than 87

Test this question on the original dataset D: rows 0–9, with 6 E and 4 U. The parent Gini is $12/25 = 0.48$. This is an alternative root trial, not an added level of the tree.

Branch	Rows and observed labels	Count
Yes	0 E, 1 U, 2 E, 4 U, 5 E, 6 U, 7 E, 8 U	8: 4 E / 4 U
No	3 E, 9 E	2: 2 E / 0 U

$$G_Y = 1 - (\frac{4}{8})^2 - (\frac{4}{8})^2 = \frac{1}{2} = 0.5000,$$
$$G_N = 1 - (\frac{2}{2})^2 - (\frac{0}{2})^2 = 0 = 0.0000,$$
$$J_G = \frac{8}{10}(\frac{1}{2}) + \frac{2}{10}(0) = \frac{2}{5} = 0.4000,$$
$$\Delta G = \frac{12}{25} - \frac{2}{5} = \frac{2}{25} = 0.0800.$$



This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

The remaining midpoint questions use the same calculation. Their final scores are:

Grade question	Weighted child Gini
Grade > 88.5	10/21 = 0.4762
Grade > 89.5	9/20 = 0.4500
Grade > 91	2/5 = 0.4000
Grade > 92.5	7/15 = 0.4667
Grade > 93.5	44/105 = 0.4190
Grade > 94.5	19/40 = 0.4750
Grade > 96.5	2/5 = 0.4000

3D Choose the best root and recurse

Compare the best numerical threshold with the categorical alternatives. Every score below uses the same ten-row parent. The smallest weighted Gini wins.

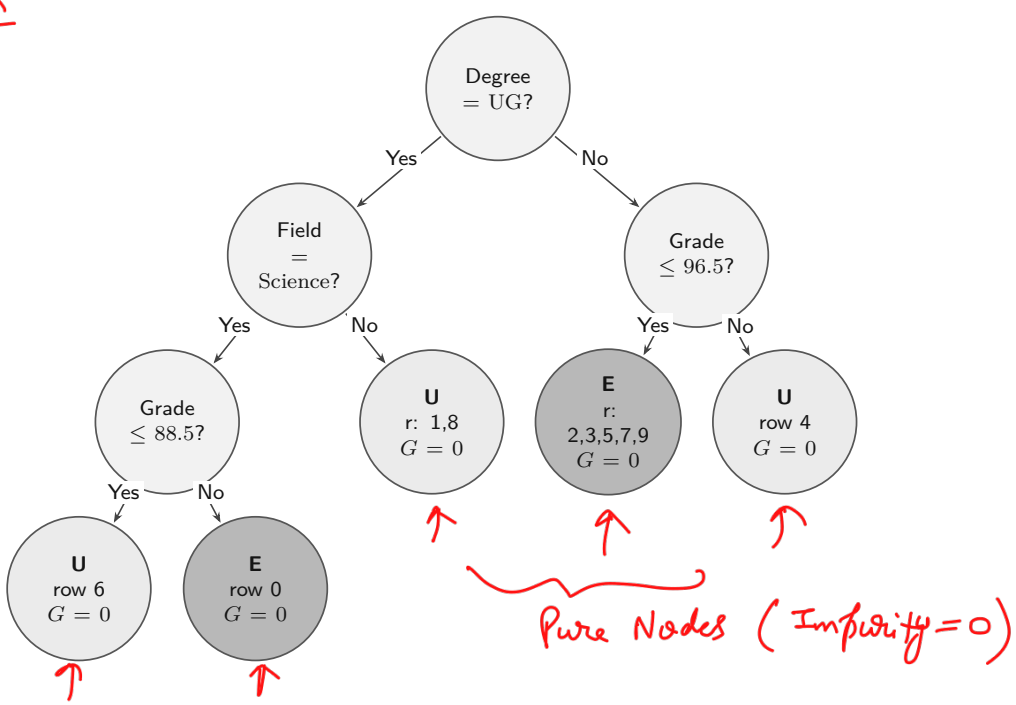
Candidate question	Weighted child Gini
Field = Science (or reversed Arts)	2/5 = 0.4000
Degree = UG	19/60 = 0.3167
Degree = PG	10/21 = 0.4762
Degree = PhD	12/35 = 0.3429
Best Grade thresholds: 87, 91 or 96.5	2/5 = 0.4000

Degree = UG is the selected root. It sends rows 0,1,6,8 to Yes and rows 2,3,4,5,7,9 to No. Its weighted Gini is 19/60 = 0.3167. Both children are still mixed: UG has 1 E and 3 U; non-UG has 5 E and 1 U. Apply the same search recursively inside each child, using only its own rows. For equal scores in this worked example, prefer Field, then Degree, then Grade; for tied Grade thresholds, prefer the lower threshold. The resulting complete tree follows.

3E Final classification tree and prediction regions

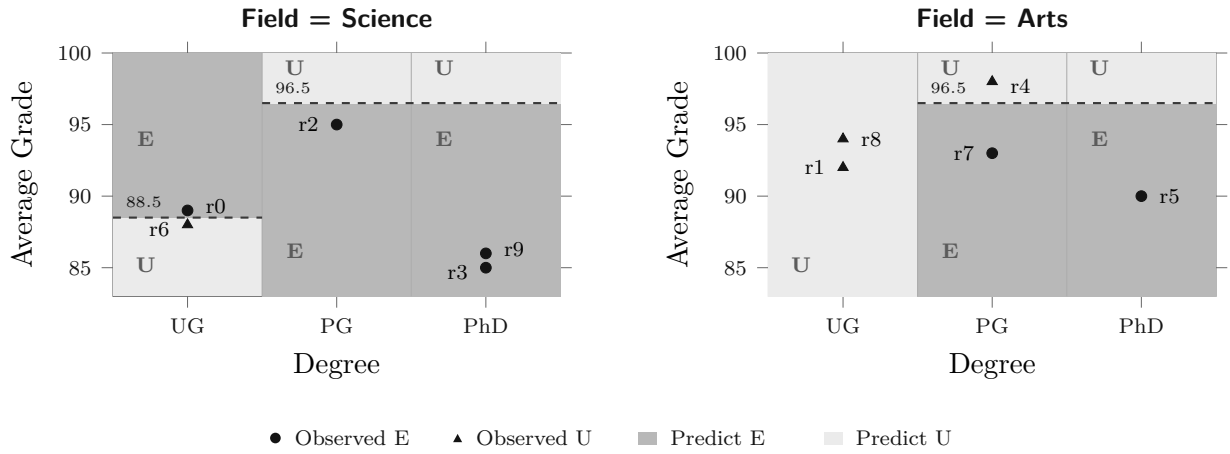
Repeating the same split-selection rule within each child gives the following tree. Every leaf is pure, so the final weighted leaf Gini reaches its minimum, **0**.

Classification



Final employment tree: five pure leaves and maximum depth three.

Final training impurity: $G_{\text{tree}} = \sum_\ell (n_\ell/10) G_\ell = 0.$



The complete classification tree as prediction regions. Each panel holds Field fixed, so the panels together represent all three inputs. Degree positions are categorical labels, not numerical distances. Dashed lines show the learned Grade thresholds; r0–r9 identify the training rows. All ten rows fall in regions matching their observed outcomes.

Read a path: UG–Science–Grade 89 predicts E. PG–Arts–Grade 98 predicts U. Zero training impurity describes these ten rows; future performance still needs new data.

3F Gini impurity intuition using fruit examples

HOOK

Why does $1 - \sum_k p_k^2$ measure mixture? Select an actual fruit at random, return it to the basket, then draw again independently and use the second fruit's class as a guess. Gini is the probability that the two class labels differ.

Three oranges and two apples

There are five equally likely choices for each independent draw, so the table contains $5 \times 5 = 25$ equally likely ordered pairs. Each cell has probability 1/25.

Distinct

O → 3

A → 2

Read the symbols: M means a **matching fruit class**; **X** means a **different fruit class**. A match compares the labels Orange/Apple, not the identity of the fruit. For example, O1 and O2 still give M.

Actual fruit (columns)

Guess ↓	O1	O2	O3	A1	A2
O1	M	M	M	X	X
O2	M	M	M	X	X
O3	M	M	M	X	X
A1	X	X	X	M	M
A2	X	X	X	M	M

Labels: O1–O3 = the three oranges; A1–A2 = the two apples. Rows represent the independent guess draw.
Light grey M: $3 \times 3 + 2 \times 2 = 13$ matches.
Dark grey X: $3 \times 2 + 2 \times 3 = 12$ mismatches.

$$P(\text{match}) = \left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 = \frac{13}{25},$$

$$G = P(\text{mismatch}) = 1 - \frac{13}{25} = \frac{12}{25} = 0.48.$$

Five equally frequent fruit classes

For five equally frequent fruit classes—orange, mango, grape, banana and cranberry—each proportion is 1/5:

$$G = 1 - 5\left(\frac{1}{5}\right)^2 = \frac{4}{5} = 0.8.$$

Again, the actual class and the guessed class are independent. The 25 class pairs each have probability $(1/5)(1/5) = 1/25$. Here **M** still means a matching class and **X** a mismatch; only the diagonal cells match.

Actual class (columns)

Guess ↓	O	Ma	G	B	C
O	M	X	X	X	X
Ma	X	M	X	X	X
G	X	X	M	X	X
B	X	X	X	M	X
C	X	X	X	X	M

Labels: O = Orange; Ma = Mango; G = Grape; B = Banana; C = Cranberry.
Grey M: 5 matches on the diagonal.
White X: 20 mismatches off the diagonal.

$$P(\text{match}) = \frac{5}{25} = \frac{1}{5},$$

$$G = P(\text{mismatch}) = \frac{20}{25} = \frac{4}{5} = 0.8.$$

A pure basket has Gini 0. With K classes, the maximum is $1 - 1/K$; with two classes, it is 0.5.

KEY INSIGHT

Gini measures independent random-label mismatch. It is not the error rate of a majority-class prediction: always predicting Orange in the first basket gives error $2/5 = 0.4$, while its Gini is 0.48.

Practice P1 Complete CART classification dry run

PRACTICE P1 – WHICH QUESTION SHOULD THE TREE ASK?

A training table records a mentoring programme and hours of preparation. The target is Selection: 1 = Selected, 0 = Not Selected. This is a new dataset; use every row and every valid candidate below.

PRACTICE P1 – WHICH QUESTION SHOULD THE TREE ASK? (continued)

Row	Hours	Mentor	Selection y
A	1	No	0
B	2	Yes	1
C	3	No	0
D	4	Yes	1
E	5	Yes	0
F	6	No	0

(a) Begin at the parent. Count the target labels and calculate the parent Gini. Then generate all distinct root questions. Mentor has two categories; Hours has six distinct ordered values.

(b) Compare every candidate. For each question, write the Yes and No class counts as 1/0, calculate both child Ginis, then calculate $J_G = (n_Y/6)G_Y + (n_N/6)G_N$. Show one full calculation beneath the table.

Yes question	Yes 1/0	No 1/0	G_Y	G_N	J_G
Mentor = Yes	_____	_____	_____	_____	_____
Hours $\leq$ 1.5?	_____	_____	_____	_____	_____
Hours $\leq$ 2.5?	_____	_____	_____	_____	_____
Hours $\leq$ 3.5?	_____	_____	_____	_____	_____
Hours $\leq$ 4.5?	_____	_____	_____	_____	_____
Hours $\leq$ 5.5?	_____	_____	_____	_____	_____

(c) Challenge a statement. A student says, "The Hours split at 1.5 has a pure Yes child, so it must beat the Mentor split." Decide whether the conclusion follows, and justify it using the whole split.

Practice P1 Recurse complete the tree and predict

PRACTICE P1 – CONTINUE THE SAME DRY RUN

(d) Recurse inside the mixed child. Use only the rows reaching Mentor = Yes. List their Hours values and labels, then score every valid local threshold. Explain why Mentor is no longer a valid split in this child.

Yes question	Yes 1/0	No 1/0	G_Y	G_N	J_G
Hours $\leq$ 3?	_____	_____	_____	_____	_____
Hours $\leq$ 4.5?	_____	_____	_____	_____	_____

PRACTICE P1 – CONTINUE THE SAME DRY RUN (continued)

(e) Complete the tree. Draw all questions, label Yes/No branches and write the row IDs and prediction in each leaf. State the number of leaves and maximum depth.

(f) Predict and explain the path. A new row has Mentor = Yes and Hours = 4.2. Another has Mentor = No and the same Hours = 4.2. Trace both paths and explain why their predictions differ.

4 CART regression predict a numerical Test Score

HOOK

The employment example predicted a categorical outcome. Now the last column is a numerical Test Score. We still ask Yes/No feature questions; each child contains a list of scores.

Consider the following eight-row dataset. Subject and Grade level are categorical inputs; Hours studied is numerical. Test Score is the target.

Row	Subject	Grade level	Hours studied	Test Score
0	Math	Freshman	4	59
1	Physics	Freshman	1	82
2	Physics	Freshman	4	81
3	Math	Junior	6	60
4	Physics	Sophomore	1	73
5	Physics	Junior	3	85
6	Physics	Junior	4	61
7	Physics	Freshman	9	78

4A Regression impurity and the split formulas

The target is numerical. Squared-error CART measures impurity using the mean squared deviation from a node's target mean, with denominator n. For the complete eight-row dataset, the final parent values are

$$\bar{y}_P = 72.375, \qquad \mathrm{MSE}_P = \frac{6559}{64} = 102.4844.$$

Variance

Use these formulas to score the candidate splits:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

$$J = \frac{n_Y}{n}\mathrm{MSE}_Y + \frac{n_N}{n}\mathrm{MSE}_N, \qquad \Delta = \mathrm{MSE}_P - J.$$

Each child uses its own target mean. Smaller weighted child MSE gives a better split; a final leaf predicts the mean of its training targets.

For exact arithmetic, if $\bar{y} = a/b$, then $y_i - \bar{y} = (by_i - a)/b$ and its square is $(by_i - a)^2/b^2$. This explains the common denominators used in the following examples.

4B Categorical example Subject equals Math

Root trial only: Start again with all eight rows. Apply this question independently; no earlier candidate has been accepted.

Yes (Y) rows $\{0, 3\}$: targets $[59, 60]$.

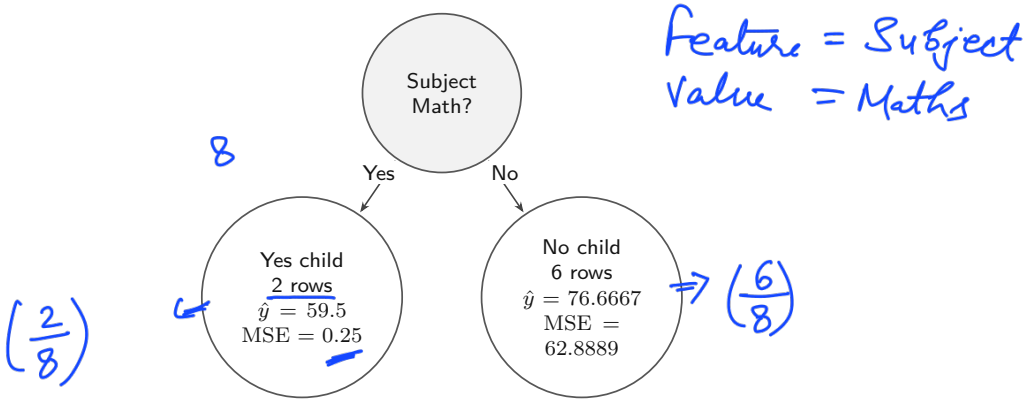
$$\mu_Y = \frac{59+60}{2} = \frac{119}{2} = 59.5,$$
$$\mathrm{MSE}_Y = \frac{(59-59.5)^2 + (60-59.5)^2}{2} = \frac{0.25+0.25}{2} = \frac{1}{4} = 0.25.$$

No (N) rows $\{1, 2, 4, 5, 6, 7\}$: targets $[82, 81, 73, 85, 61, 78]$.

$$\mu_N = \frac{82+81+73+85+61+78}{6} = \frac{230}{3} = 76.6667,$$
$$\mathrm{MSE}_N = \frac{1}{6}\Big[(82-\tfrac{230}{3})^2 + (81-\tfrac{230}{3})^2 + (73-\tfrac{230}{3})^2 + (85-\tfrac{230}{3})^2 + (61-\tfrac{230}{3})^2 + (78-\tfrac{230}{3})^2\Big]$$
$$= \frac{16^2 + 13^2 + (-11)^2 + 25^2 + (-47)^2 + 4^2}{6 \cdot 3^2} = \frac{566}{9} = 62.8889.$$
$$J = \frac{2}{8}\left(\frac{1}{4}\right) + \frac{6}{8}\left(\frac{566}{9}\right) = \frac{2267}{48} = 47.2292.$$

Overall weighted variance

Decrease from this parent: $\Delta = \frac{6559}{64} - \frac{2267}{48} = \frac{10609}{192} = 55.2552.$



This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

Subject has only Math and Physics. Asking Subject = Physics reverses these two children, so its weighted MSE is also $2267/48 = 47.2292$.

The categorical Grade questions use the same process; their final weighted MSEs are:

Grade question	Weighted child MSE
Grade = Freshman	$3059/32 = 95.5938$
Grade = Junior	$2827/30 = 94.2333$
Grade = Sophomore	$717/7 = 102.4286$

4C Numerical inputs sort Hours and generate thresholds

Subject and Grade level use category-membership questions. Hours studied uses a threshold. Keep all eight rows, sort their Hours values, and carry each row's target with it.

Position	Row	Hours	Test Score
1	1	1	82
2	4	1	73
3	5	3	85
4	0	4	59
5	2	4	81
6	6	4	61
7	3	6	60
8	7	9	78

The distinct Hours values are $1, 3, 4, 6, 9$. Candidate thresholds lie halfway between adjacent distinct values:

Adjacent distinct values	Midpoint	Trial question
1 and 3	$(1+3)/2 = 2$	Hours ≤ 2?
3 and 4	$(3+4)/2 = 3.5$	Hours ≤ 3.5?
4 and 6	$(4+6)/2 = 5$	Hours ≤ 5?
6 and 9	$(6+9)/2 = 7.5$	Hours ≤ 7.5?

KEY INSIGHT

Rows 0, 2 and 6 all have Hours = 4. A threshold on Hours cannot put these equal inputs into different children, even though their scores are 59, 81 and 61.

Score each midpoint using the same weighted MSE rule. The following example works through Hours at or below 2; the remaining results are summarised in a table. Use exactly the same weighted MSE rule as for categories.

TAKEAWAY

Generate four numerical candidates from distinct Hours values. The scores decide which candidate to keep; sorting alone does not select the split.

4C Numerical example Hours at or below 2

Root trial only: Start again with all eight rows. Apply this question independently; no earlier candidate has been accepted.

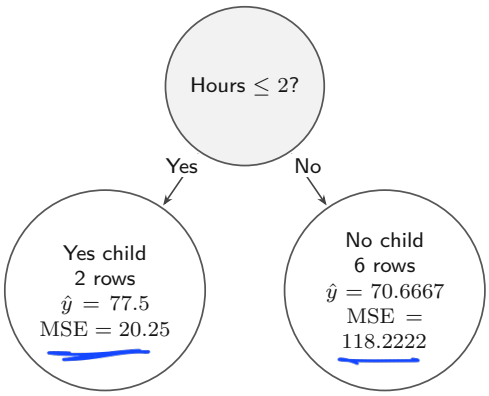
Yes (Y) rows $\{1, 4\}$: targets $[82, 73]$.

$$\mu_Y = \frac{82+73}{2} = \frac{155}{2} = 77.5,$$
$$\text{MSE}_Y = \frac{(82-77.5)^2 + (73-77.5)^2}{2}$$
$$= \frac{20.25 + 20.25}{2} = \frac{81}{4} = 20.25.$$

No (N) rows $\{0,2,3,5,6,7\}$: targets $[59,81,60,85,61,78]$.

$$\mu_N = \frac{59+81+60+85+61+78}{6} = \frac{212}{3} = 70.6667,$$
$$\begin{aligned}\mathrm{MSE}_N &= \frac{1}{6}\Big[(59-\tfrac{212}{3})^2+(81-\tfrac{212}{3})^2+(60-\tfrac{212}{3})^2 \\ &\quad +(85-\tfrac{212}{3})^2+(61-\tfrac{212}{3})^2+(78-\tfrac{212}{3})^2\Big] \\ &= \frac{(-35)^2+31^2+(-32)^2+43^2+(-29)^2+22^2}{6\cdot 3^2} = \frac{1064}{9} = 118.2222.\end{aligned}$$
$$J = \frac{2}{8}\Big(\frac{81}{4}\Big)+\frac{6}{8}\Big(\frac{1064}{9}\Big) = \frac{4499}{48} = 93.7292.$$

Decrease from this parent: $\Delta = \frac{6559}{64} - \frac{4499}{48} = \frac{1681}{192} = 8.7552.$



This is a candidate question. Compare its weighted child impurity with the other candidates before selecting a root.

The remaining Hours questions follow the same calculation. Their final weighted MSEs are:

Hours question	Weighted child MSE
Hours ≤ 3.5	$338/5 = 67.6000$
Hours ≤ 5	$1579/16 = 98.6875$
Hours ≤ 7.5	$2743/28 = 97.9643$

4D Select the regression root and recurse

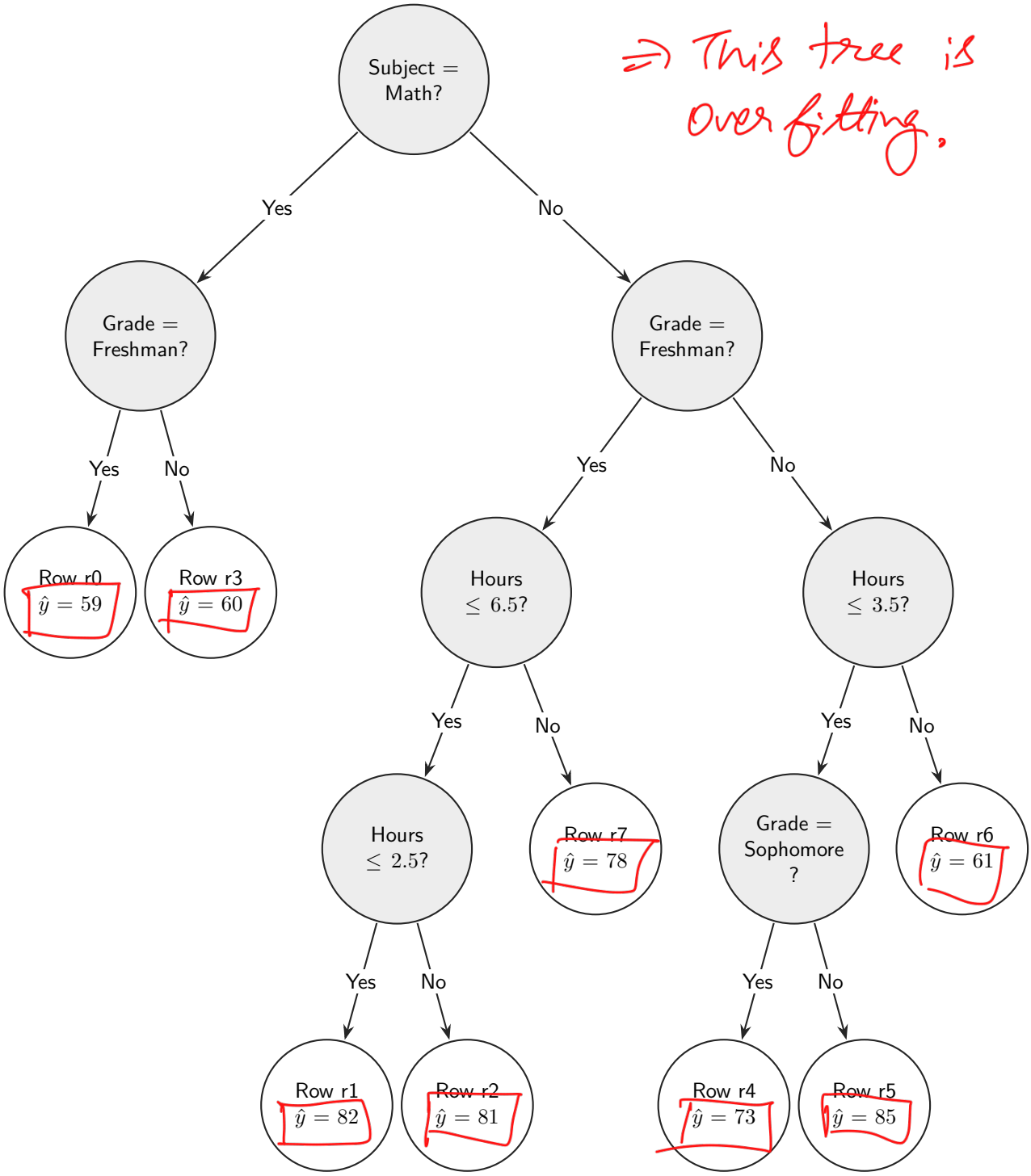
Compare the strongest question from each feature. All scores use the original eight-row dataset.

Feature	Best question	Weighted child MSE
Subject	Subject = Math	$2267/48 = 47.2292$
Grade	Grade = Junior	$2827/30 = 94.2333$
Hours	Hours ≤ 3.5	$338/5 = 67.6000$

Subject = Math is selected as the root. The reversed Physics question ties it; keep Math as the orientation because it was tested first. At this first split, the current leaf predictions are 59.5 for Math and 76.6667 for Physics. Repeat the same search inside each child. For tied questions, use Grade before Hours in this worked example; for tied Hours thresholds, use the smaller threshold. The complete resulting tree follows.

4E Complete regression tree

Follow one question at a time from the root to a circular leaf. The leaf shows the training row and its predicted Test Score.

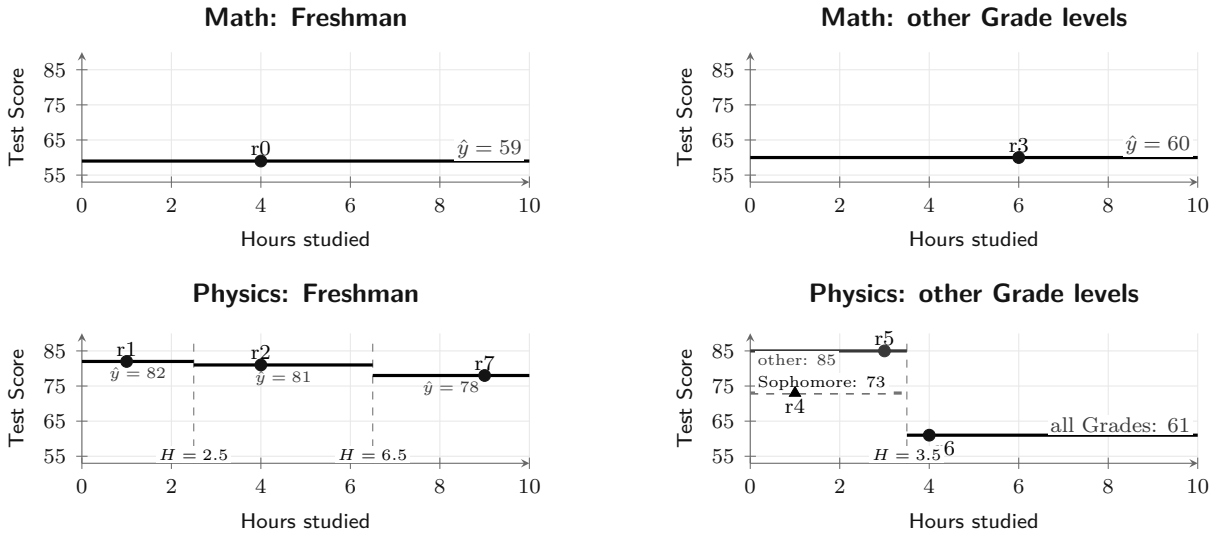


⇒ This tree is overfitting.

9

Eight leaves; maximum depth four. Every leaf predicts its target mean. Here each leaf contains one training row, so the final training MSE is 0.

4F Read the final regression predictions



Final numerical predictions from the complete tree. Each panel fixes the Subject and Grade branch; Hours is an input and Test Score is the numerical target. Points mark observed training scores, solid segments show leaf predictions, and dashed lines mark Hours thresholds active in that branch only. The final panel shows its additional Grade question: the dashed segment predicts 73 for Sophomore and the solid upper segment predicts 85 for other non-Freshman Grades, until Hours exceeds 3.5. The lower solid segment predicts 61 after that threshold. All eight observations now lie on their predicted segments.

Zero training MSE verifies this fitted example; it does not guarantee accurate unseen scores.

Practice P2 Complete CART regression dry run

PRACTICE P2 – GROW A TREE USING SQUARED ERROR

Predict a numerical test score using Course and Hours. Use squared-error CART. Let the tree grow until the targets in each leaf are constant; no depth or leaf-size limit is imposed in this paper dry run.

Row	Hours	Course	Test score y
A	1	A	10
B	2	B	50
C	3	A	12
D	4	B	50
E	5	A	30
F	6	B	50

(a) Start with one leaf. Compute the parent mean and MSE. Show the average squared deviation from the mean.

(b) Score all root questions. Complete the child statistics for each candidate. In each child use its own mean; calculate

$$J_{\text{MSE}} = \frac{n_Y}{6}\text{MSE}_Y + \frac{n_N}{6}\text{MSE}_N.$$

PRACTICE P2 – GROW A TREE USING SQUARED ERROR (continued)

Yes question	Yes child	No child	J_{MSE}
Course = A	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____
Hours ≤ 1.5?	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____
Hours ≤ 2.5?	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____
Hours ≤ 3.5?	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____
Hours ≤ 4.5?	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____
Hours ≤ 5.5?	$n =$ ___, $\mu =$ ____ MSE = ____	$n =$ ___, $\mu =$ ____ MSE = ____	_____

(c) Choose the root and justify it. Identify the best candidate and show its weighted calculation. Explain why taking an unweighted average of child MSEs is not generally valid.

Practice P2 Recurse calculate leaves and predict

PRACTICE P2 – FINISH THE SAME REGRESSION TREE

(d) Restart the search inside each child. Course = B contains three scores equal to 50. Course = A contains $(H, y) = (1, 10), (3, 12), (5, 30)$. Calculate the two valid local threshold scores and choose one.

Local split	Yes mean / MSE	No mean / MSE	J_{MSE}
Hours ≤ 2?	_____	_____	_____
Hours ≤ 4?	_____	_____	_____

(e) Complete the recursion. The Hours ≤ 4 child still has scores 10 and 12. Find its remaining candidate, calculate its score, and complete the entire tree with numerical leaf predictions.

PRACTICE P2 – FINISH THE SAME REGRESSION TREE (continued)

(f) Predict and interpret. Trace a new row with Course = A and Hours = 4.8. Would moving its Hours to 6 force the prediction above every training score? Does zero training MSE establish good performance on unseen data?

(g) Explain the leaf rule. Why do these regression leaves use means instead of the most frequent score?

Practice P3 Comprehensive challenge

PRACTICE P3 · CONCEPTS AND STATEMENTS

(a) True or False. Justify each answer; correct the false statements.
(i) A node with 6 employed and 4 unemployed people has Gini 0.48, so its majority-class prediction misclassifies 48% of those people.

(ii) Coding Placement as 0 and 1 makes prediction of the Placement category a regression task.

(iii) A regression tree with zero training MSE must also predict every unseen student score perfectly.

(iv) Two observations reaching the same regression leaf receive the same prediction, even when their feature values differ.

(b) Match the columns. Match each situation to its interpretation, then justify any one match.

Situation	Interpretation
(1) Choose among classification candidates.	(A) Give every row equal weight through child counts.
(2) Compare children of sizes 1 and 9.	(B) Use the mean of the targets in that leaf.
(3) Predict a new score after routing it.	(C) Choose the lowest weighted child Gini.
(4) Recurse inside the UG child.	(D) Search using only the rows that reached this node.

(c) Challenge a choice. At a ten-row parent, candidate A has a pure one-row child and a nine-row child with Gini 40/81. Candidate B has child sizes 4 and 6, with Ginis 3/8 and 5/18. A classmate selects A because "one child is completely pure." Calculate both weighted scores and explain which candidate should be chosen.

5 Advantages and disadvantages of decision trees

We have now built trees by trying feature questions, routing whole rows and comparing the two children. In classification, the calculation measured class mixture; in regression, it measured score spread. The same sequence of questions can therefore support both prediction tasks.

Advantages

Strength	Connection to our examples
Readable decisions	A small tree explains a prediction through a short sequence of if–else answers. We can follow the exact path to a leaf.
Classification and regression	The employment tree predicts a class; the score tree predicts a number. The routing process is shared.
Nonlinear rules and interactions	A question can appear only inside an earlier branch. Its effect can depend on previous answers.
Little scaling preparation	Numerical inputs usually need no standardization. Trees can use numerical thresholds and categorical partitions, with suitable data representation.
Fast prediction paths	Prediction checks only the nodes on one root-to-leaf path. The work follows the path's depth, rather than visiting every node.

Disadvantages

Limitation	What it means
Excessive growth	A deep tree can memorize noise and become difficult to explain. A perfect fit to the training rows need not carry over to new rows.
Instability	Small changes to the data can change a preferred split, which can also change later branches and predictions.
Dominant classes	When one class is much more frequent, a tree may favor it and make poor predictions for the less frequent class.
Data format	Practical software may require categorical encoding and particular missing-value handling; these depend on the implementation.

TAKEAWAY

A decision tree turns successive local questions into a prediction path. Its strongest benefit is a clear connection between the rows, the splits and the final prediction; its main risk is growing a rule that fits the training data too closely.